

Week 5: Systems of Linear Equations

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Week 5 Quiz

5.1: Systems of linear equations: elimination (standard form set-up, clean arithmetic)

5.2: Systems: applications & modeling (set up from word problems)

5.3: Systems: substitution; classify (one/none/infinite)

Quick Reference: Equations, Definitions, & Formulas

Slope-Intercept Form

$$y = mx + b$$

where m is the slope and b is the y-intercept.

Point-Slope Form

$$y - y_1 = m(x - x_1)$$

where m is the slope and (x_1, y_1) is a point on the line.

Slope Formula

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

The slope between two points (x_1, y_1) and (x_2, y_2) .

Types of Systems

- **One solution:** Lines intersect at one point (different slopes)
- **No solutions:** Lines are parallel (same slope, different y-intercepts)
- **Infinitely many solutions:** Lines are the same (same slope, same y-intercept)

Solution Methods

- **Substitution:** Solve one equation for a variable, substitute into the other
- **Elimination:** Add or subtract equations to eliminate a variable
- **Graphing:** Graph both lines and find the intersection point

Week 5 Quiz: Systems of Linear Equations

Part A: Writing Equations of Lines

Instructions: Find the equation of each line in slope-intercept form:

1. Find the equation of the line with a slope of 3 and a y-intercept of -2 .
2. Find the equation of the line with a slope of $\frac{2}{3}$ and contains the point $(6, 7)$.
3. Find the equation of the line through the points $(3, 5)$ and $(-1, 9)$.

Part B: Graphing Systems

Instructions: Graph each line. Classify the system (one solution / no solutions / infinitely many solutions). Find the solution to the system.

1. $y = \frac{1}{2}x - 3$
 $y = -x + 3$
2. $y = \frac{1}{3}x + 1$
 $x - 3y = -6$

Part C: Solving Systems Algebraically

Instructions: Solve the following systems of equations. Show all work.

1. $y = x - 6$
 $y = -x + 3$
2. $y = -\frac{3}{2}x + 9$
 $3x + 2y = 18$
3. $3y = -2x + 7$
 $-2y = 2x - 3$

Part D: Word Problems

Instructions: Define variables (with units), determine the two equations in the system, solve the system, and state the final answer (with units).

1. Two gyms charge an enrollment fee plus an additional cost per month. Let m be months and C be total cost (in \$).
Gym A charges \$35 per month with a \$100 enrollment fee.
Gym B charges \$45 per month with a \$50 enrollment fee.
Write an equation for the cost, C (in \$), with respect to m (in months), for both gyms.
What is the number of months where both gyms have the same total cost (if it exists)?
2. A school sold hats for a fundraiser in two sizes: small hats and large hats.
There were 50 hats sold in total.
Each small hat cost \$12.
Each large hat cost \$7.
The total for all the hats sold was \$490.
How many of each type of hat was sold?

Solution Guide

Part A: Writing Equations of Lines

1. Find the equation of the line with a slope of 3 and a y-intercept of -2 .

Solution:

Use the slope-intercept form $y = mx + b$

$$m = 3, b = -2$$

$$y = 3x - 2$$

Answer: $y = 3x - 2$

2. Find the equation of the line with a slope of $\frac{2}{3}$ and contains the point $(6, 7)$.

Solution:

Use the point-slope form $y - y_1 = m(x - x_1)$

$$m = \frac{2}{3}, (x_1, y_1) = (6, 7)$$

$$y - 7 = \frac{2}{3}(x - 6)$$

$$y - 7 = \frac{2}{3}x - 4$$

$$y = \frac{2}{3}x + 3$$

Answer: $y = \frac{2}{3}x + 3$

3. Find the equation of the line through the points $(3, 5)$ and $(-1, 9)$.

Solution:

First find the slope:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{9 - 5}{-1 - 3} = \frac{4}{-4} = -1$$

Now use point-slope form with $(-1, 9)$:

$$y - 9 = -1(x - (-1))$$

$$y - 9 = -x - 1$$

$$y = -x + 8$$

Answer: $y = -x + 8$

Part B: Graphing Systems

1. $y = \frac{1}{2}x - 3$ and $y = -x + 3$

Solution:

Set the equations equal:

$$\frac{1}{2}x - 3 = -x + 3$$

$$\frac{3}{2}x = 6$$

$$x = 4$$

Substitute $x = 4$ into $y = -x + 3$:

$$y = -4 + 3 = -1$$

Answer: One solution, $(4, -1)$

2. $y = \frac{1}{3}x + 1$ and $x - 3y = -6$

Solution:

Use substitution method:

$$x - 3\left(\frac{1}{3}x + 1\right) = -6$$

$$x - x - 3 = -6$$

$$-3 = -6$$

This is a false statement, so the lines are parallel.

Answer: No solutions (parallel lines)

Part C: Solving Systems Algebraically

1. $y = x - 6$ and $y = -x + 3$

Solution:

Use substitution method:

$$x - 6 = -x + 3$$

$$2x = 9$$

$$x = \frac{9}{2}$$

Substitute into $y = x - 6$:

$$y = \frac{9}{2} - 6 = \frac{9}{2} - \frac{12}{2} = -\frac{3}{2}$$

Answer: $(\frac{9}{2}, -\frac{3}{2}) = (4.5, -1.5)$

2. $y = -\frac{3}{2}x + 9$ and $3x + 2y = 18$

Solution:

Use substitution method:

$$3x + 2(-\frac{3}{2}x + 9) = 18$$

$$3x - 3x + 18 = 18$$

$$18 = 18$$

This is always true, so the lines are the same.

Answer: Infinitely many solutions (same line)

3. $3y = -2x + 7$ and $-2y = 2x - 3$

Solution:

Use elimination method (add the equations):

$$3y = -2x + 7$$

$$-2y = 2x - 3$$

$$y = 4$$

Substitute $y = 4$ into $-2y = 2x - 3$:

$$-2(4) = 2x - 3$$

$$-8 = 2x - 3$$

$$2x = -5$$

$$x = -\frac{5}{2}$$

Answer: $(-\frac{5}{2}, 4)$

Part D: Word Problems

1. Two gyms charge an enrollment fee plus an additional cost per month.

Solution:

Variables: Let m = months, C = total cost (in \$)

Gym A: $C = 35m + 100$

Gym B: $C = 45m + 50$

Set equal and solve:

$$35m + 100 = 45m + 50$$

$$50 = 10m$$

$$m = 5$$

Answer: 5 months

2. A school sold hats for a fundraiser in two sizes.

Solution:

Variables: Let S = number of small hats, L = number of large hats

Equations:

$$S + L = 50 \text{ (total hats)}$$

$$12S + 7L = 490 \text{ (total cost)}$$

Use substitution method:

$$L = 50 - S$$

$$12S + 7(50 - S) = 490$$

$$12S + 350 - 7S = 490$$

$$5S = 140$$

$$S = 28$$

Substitute back:

$$L = 50 - 28 = 22$$

Answer: 28 small hats & 22 large hats
